

B. Tech Degree VII Semester (Supplementary) Examination July 2010

EB/EC/EI 705 (D) MECHATRONICS (2006 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART AAnswer ALL questions

(8 x 5 =40)

- I. a. What is Mechatronics? Discuss on the scope of Mechatronics.
 b. Define five parameters used to represent the performance of transducers. Suggest example.
 c. Explain the need for signal conditioning. Illustrate with example.
 d. Explain the principle of a pilot operated valve.
 e. Explain the principle of operation of a Cam.
 f. Define 4 parameters to measure the performance of a second order system
 g. Explain what is MEMS and its importance in Mechatronics.
 h. Explain the role of intelligent controllers in Mechatronics.

PART B

(4 x 15 = 60)

- II. a. You are offered a choice of an incremental shaft encoder or an absolute shaft encoder for the measurement of an angular displacement. What is the typical difference between the results that can be obtained by these methods? (8)
 b. What is the resolution of an analog to digital converter with a word length of 12 bits and an analogue signal input of 100V? (7)

OR

- III. a. Suggest a sensor that could be used as part of a control system to determine the difference in levels between liquids in two containers. The output is to provide an electrical signal for the control system. Explain the principle of operation of such a sensor. (8)
 b. Design an operational amplifier circuit that can be used to produce an output that ranges from 0 to -5V when the input goes from 0 to 100mV. (7)

- IV. a. Describe valve symbols, with suitable example. (7)
 b. Describe the operation of the Four-bar chain with neat diagrams. (8)

OR

- V. a. Describe the features of any two solid-state switches. (7)
 b. Explain what is pulse width modulation and its use in speed control of DC motors. (8)

- VI. a. Explain the building blocks used for mechanical and electrical systems. Relate their mathematical models. (8)
 b. Derive the mathematical model of an armature controlled DC motor. (7)

OR

- VII. a. What will be the output, if a system with the transfer function $s/(s+3)(s+4)$ is subjected to a unit step input at time $t=0$? (8)
 b. Explain, in detail, any one manufacturing process of MEMS. (7)

- VIII. a. Describe and compare the characteristics of proportional plus integral control and proportional plus derivative control. (7)
 b. Discuss on different types of Robot configurations. (8)

OR

- IX. a. Differentiate between programmable logic controller (PLC) & computer. Suggest one application of PLC in mechatronics, in detail. (7)
 b. Write short notes on (i) FMS and (ii) FMC. (8)